

# Effects of Customisation of AI Shopping Assistant on VR Retail

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**Figure 1: The primary VR experience setting.** On the left (a) is where the participant begins the experience. In the middle (b) is the location of the shopping assistant, and on the right (c) is the location of the cashier, where users go to check out.

## Abstract

VR Retail is a promising extension of the increasingly popular online shopping and can provide the convenience of online stores with the immersion of physical stores. However, little has been done to explore how to improve the VR retail shopping experience. This paper investigated the potential impacts of customizing virtual shopping assistants on VR shopping experiences. A total of 27 participants were recruited for a within-subject study to undergo shopping experiences with either a customised or non-customised shopping assistant. This study found that users experienced higher social presence and spent more in the condition with a customised shopping assistant. The results of this study reveal avenues in which VR Retail can be improved upon for adoption in the mainstream market.

## CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction paradigms**; • **Virtual Reality**;

## Keywords

Customisation, Virtual Reality Retail, Shopping Assistant

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## 1 Introduction

Improvements in Virtual Reality (VR) technology have led to the successful deployment of VR experiences in health, education, and entertainment [1–4]. The popularity of online shopping has also inspired work into the development of VR retail experiences [5–7]. Research suggests that VR Retail can better fulfill hedonic shopping



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motivations than online shopping [8], increase spending, and is more efficient than shopping in physical stores [9, 10].

Current literature on avatar customisation mainly focuses on user-controlled avatars. In [11–13], research is done on the impacts of user-controlled avatars on the VR experience; however, little research is done on non-user avatars. Customisation has been shown through previous literature as a technique that generally enhances VR experiences through a large range of factors [1, 14, 15]. Though there is a lack of focus on the customisation of non-player characters. This paper looks at the potential enhancements to the VR shopping experience that can be brought about through the customisation of a VR shopping assistant. Shopping assistants are a common presence in physical stores and have been widely used in E-commerce (e.g., IKEA's Billie, Amazon's Alexa, and Apple's Siri); however, they are underexplored in VR retail experience.

In this paper, we conducted a within-subjects study with 27 participants to shop in VR retail with (1) a customised and (2) a non-customised assistant. Our results show an increase in social presence and spending when allowing users to customize their shopping assistants, indicating a positive effect on the efficacy of VR Retail experiences. The contributions of this paper are: (1) being the first investigation of customizing AI-enabled virtual avatars in VR Retail. (2) We demonstrate that customizing virtual shopping assistants in VR retail enhances social presence and increases consumer spending.

## 2 Related Works

### 2.1 Shopping Assistant

In “The Media Equation,” Reeves and Nass [16] state that when interacting with computers as social actors (CASA), “it is belief, not disbelief, that is automatic.” This means that unless otherwise presented with reason to doubt the humanness of the computer, users will tend to treat them as human.

Virtual assistants are one such example of CASA and are commonly used within retail (e.g., Alexa, Siri, IKEA's Billie). There is a focus on perceived competency and perceived anthropomorphism among shopping assistant studies [17, 18]. Waytz et al. [19] explain that Anthropomorphism is when people attribute human characteristics to nonhumans, such as the capacity for rational thought or conscious feeling. They also found that anthropomorphism of a given agent increased trust in the agent. The link between anthropomorphism and trust is also supported by [20]. In the context of a shopping assistant, trust is particularly important to enhance the VR shopping experience, as Gefen [21] found that trust and familiarity are strongly related to purchase within E-commerce. It is also mentioned that familiarity is a precursor to trust, which is also supported by [22]. Therefore, we plan to use an anthropomorphic avatar as a shopping assistant in this work.

### 2.2 Avatar Customisation

Customisation has been widely studied within retail and can enhance user experience twofold, giving the user agency within their experience or customer involvement [23] and by allowing the experience and outcomes to better suit the user or customised product [24]. Marathe and Sundar [25] also investigated the motivations

for user customisation and found that customisation is associated with a sense of control and a sense of identity.

Cheung and To [23] found that customer involvement positively relates to perceived empathy of the service agent. Though the service agent in this case was a real person, the relationship between customer involvement and empathy should still exist when interacting with a virtual service agent, as Nass and Moon [26] mention that “the computer seems to be a peer in a social interaction”. In the context of this paper, the medium is the virtual avatar rather than the computer. As for customised products, it has been shown to positively influence purchasing decisions [24, 27].

Franke et al. [28] identified that customisation of products benefits purchase intentions and that this effect is enhanced by users' ability to express their preferences. Although it referred to the customisation of products, this effect could extend to shopping experiences by allowing users to customize their shopping assistants. Sundar et al. [29] found that system-initiated personalisation had a larger effect on non-power users, while user-initiated customisation had a larger effect on power users with regard to the content quality of their news website. Power usage is categorised as “a user's motivation, efficacy, expertise, and demonstration of evolved technology use” [29] which are concepts that closely relate to technology acceptance, “the consumer's engagement in electronic exchange relationships with Web retailers” as described by Pavlou [30] which in the context of this paper is engagement with VR experiences. This suggests a relationship between the desire to engage with customisation and technology acceptance. As such, research with customisation needs to be aware of less technically minded users for more prominent effects.

Customisation has also been used for online recommendation agents, and it was found that perceived customisation increased trust, demonstrated by users adopting recommendations made by the agent [17]. The increase in trust due to perceived customisation suggests that customizing a VR shopping assistant could similarly increase trust, which directly relates to spending and anthropomorphism. Customisation within VR has also been used extensively to enhance the efficacy of VR experiences in a wide range of settings [2, 12, 14, 31, 32] and is generally focused on user embodied avatars, though there is a lack of customisation applied in VR retail spaces. Such customisations have mainly focused on the recreation of the player avatars [12, 32, 33]. For instance, Waltemate et al. [12] investigated the impacts of avatar personalisation on virtual body ownership and presence by allowing users to embody avatars that were increasingly similar to the user within VR. Radiah et al. [32] conducted a similar study by investigating users embodying avatars of varying genders and accuracy to the user. There is a lack of literature on the customisation of non-player character avatars and their effects on user perception, which we aim to explore in this study.

### 2.3 Shopping Motivations

Shopping motivations can be broken into utilitarian and hedonic shopping motivations [34]. Utilitarian shopping motivations refer to shopping strictly for need or practicality, while hedonic motivations are social and emotion-based. To understand the effects of customizing a shopping assistant on VR Retail, it is necessary to

see how the shopping experience satisfies both of these shopping motivations.

P-L To et al. [34] identified that these shopping motivations raise purchase intention; therefore, measuring the amount spent should help directly quantify the quality of the shopping experience. P-L To et al. [34] also broke down utilitarian shopping motivations of E-commerce into cost saving, convenience, selection, information availability, lack of sociality, and customised product or service. The inclusion of a shopping assistant should help fulfil these utilitarian needs.

Zhou et al. [35] found that hedonic shoppers had increased interactivity in the environment and also increased enjoyment when doing online shopping. Given that the usage of VR shopping is to provide a more interactive experience for users, VR shopping will likely influence hedonic shopping motivations. The customisation of a shopping assistant could enhance the efficacy of the VR shopping experience to fulfill hedonic needs. Arnold et al. [36] categorised 6 different types of shopping motivations which under hedonic motivations—(1) Adventure shopping: the enjoyment derived from shopping as sightseeing, (2) Social shopping: the enjoyment through socializing with others, (3) Gratification shopping: shopping to alleviate negative moods such as stress, (4) Idea shopping: the enjoyment of seeing new products, (5) Role shopping: the enjoyment of shopping for others, (6) Value shopping: the enjoyment through getting deals or high-value purchases. Most applicable to this study are (2), (3), and (6). The inclusion of a virtual assistant could enhance these motivations by giving a target for socialisation, acting as a support avatar [1], and assisting with value shopping by providing product information.

## 2.4 Telepresence, Co-Presence, and Social Presence

Lombard and Ditton [37] summarised presence across literature into 6 parts: (1) Presence as social richness, (2) Presence as realism, (3) Presence as transportation (4) Presence as immersion (5) Presence as social actor within medium (6) Presence as medium as social actor. Of relevance to the study are (1), (4), and (5), as within VR, presence can be broken down into 3 sub-categories: telepresence, co-presence, and social presence [38, 39]. Presence as immersion or Tele-presence is the user's sense of being within the virtual environment, Lombard and Ditton [37] states that telepresence "can be objectively measured by counting the number of the users' senses that are provided with input and the degree to which inputs from the physical environment are 'shut out'". This highlights the environmental aspect of the VR experience. It's a measure that focuses only on the user and is used within VR literature as a measure of the user's immersion in the VR space and the general efficacy of the VR experience to convey that the environment is real [12, 38]. Which, in the context of this paper, is the believability of the VR experience emulating a store. Nowak and Biocca [38] state that co-presence "requires that interactants feel they were able to perceive their interaction partner, and that their interaction partner actively perceived them". As such, co-presence measures the degree to which the user perceives that they are sharing space with another. It requires the existence of other avatars within the same VR space and is therefore only used in research about the

user's perception of others [1, 38, 40]. For this study, co-presence is required to measure the ability of the shopping assistant avatar to facilitate interactions with the user. Presence as social richness or social presence is described as "the extent to which a medium is perceived as sociable, warm, sensitive, personal or intimate when it is used to interact with other people," by Lombard and Ditton [37] who categorize it with Social Presence theory [41] and media richness theory [42]. However, this study involves interactions with a virtual agent as such social presence in this context also includes presence as social actor within medium, which relates to CASA [43]. Bailenson et al [39] use this aspect of social presence as a measure for the user's perception that the virtual agent is conscious and aware. Measuring social presence should provide insights into the authenticity of the interactions between the user and the shopping assistant. A higher sense of social presence should also correlate with a higher sense of perceived anthropomorphism, as they both refer to the user's perception of the shopping assistant as having conscious thought [44, 45]. Munnukka et al. [46] found that perceived anthropomorphism strongly influences the formation of social presence and that social presence led to trust within the virtual service assistant, which is also supported by [47–49].

## 2.5 Research Gaps and Research Questions

Given the lack of literature on customisation of virtual assistants and their relation to social presence. The following research questions were developed:

- RQ1: Does the users' sense of Social Presence increase when given the ability to customize their shopping assistant?
- RQ2: Does the user spend more on purchases when given the ability to customize their shopping assistant?
- RQ3: Does customisation of the shopping assistant help fulfill users' shopping motivations?

## 3 EXPERIMENT

### 3.1 Experiment Design and Apparatus

The experiment followed a within-subject design with two conditions (1. Non-customised shopping assistant and 2. Customised shopping assistant) to explore the effect of customisation of the shopping assistant on VR shopping experience. We used the HTC Vive Pro as the VR device, which is connected to a PC using an NVIDIA RTX A6000 GPU. Participants used two HTC Vive controllers for VR interactions. The experiment took place in a controlled lab room. Unreal Engine 5.1 was used to develop the VR retail experience while the shopping assistant avatars were made using Metahuman. A free Unreal Engine plugin was used to record and transcribe the user's voice input. We then used the OpenAI API to process the voice inputs of the user as a prompt for ChatGPT to generate a reply with provided meta data of the store. We then used the Metahumans API to generate a corresponding audio output and facial animation for the avatar as a response to the inputs given by the participants. The study was approved by the Human Ethics Committee of University of Queensland.

### 3.2 Virtual Reality Store

The store was modelled after convenience stores, with the items and their price based on the real world. The layout consisted of the entrance where users would start and pick up their shopping basket, shelves where the items were located, the shopping assistant, which was stationary, and the checkout assistant, which did not have talking capabilities and only served to end the shopping experience. Users would shop by picking up their shopping baskets and placing items into said baskets before checking out at the cashier to end the experiment. Talking with the shopping assistant was non-mandatory for the experiment, and interactions were used as a measurement.

### 3.3 Assistant Interaction and Customisation Design

We followed [33] on customizing avatars to match the user's appearance with categorical selections: skin tone, hair colour, and gender. This matched well with the design requirements, as models must be taken from the MetaHumans presets to work with the lip sync animations. The extent of customisation for the shopping assistant is purely cosmetic, and the options were classified into gender (Female/Male), age (Old/Young), and clothing (Formal/Casual), with the additional choice of 3 different skin tones (Light/Light Brown/Dark). Choices were made by the user in a menu screen through drop-down selections, with the exception of skin tones, which were made through a slide bar. The selection made by the users was reflected in a preview image on the menu screen. The choice of categories for the customisation was made to include and distinguish the available MetaHuman presets as easily as possible while allowing users to select from a reasonable variety of assistants such that the choices felt meaningful. Only cosmetic changes were allowed to limit the scope of the study to customisation, as changing the functionality of the assistant would change the focus to variations of assistants. This is because technically demanding choices would have made the effect of customisation less evident within the non-power user groups [29]. The assistant will remain stationary in a set location and require users to approach and enter a set proximity before engaging in conversation with the assistant. To alert users that they can begin a conversation with the shopping assistant, the assistant will outline white when users are in range. The conversation will begin when users press and hold the bottom of the touchpad on the controller. A list of possible VR assistant customisation options can be seen in Figure 2.

Some example questions for the assistant were given to participants to demonstrate the functionality of the assistant. Such questions included, "how much have I spent?" to which the assistant to reply with the total cost of items within the basket or "where can I find (product)?" to which the assistant would verbally communicate the location of the product as well as highlight the location within the VR store.

### 3.4 Experiment Procedure

Participants signed the consent form before the start of the experiment. The experiment started with a short VR training session to familiarize the users with the controls and examples of assistant abilities. Once familiar with the controls, participants underwent



**Figure 2: The preset customization options of the shopping assistant from MetaHumans**

two main VR shopping sessions (1. Non-customised shopping assistant and 2. Customised shopping assistant) in random order. The main VR session had participants take a pre-test and then customize their assistant if it was the customised group, followed by the VR shopping experience and post-test measurements. After the experiment, participants took part in a semi-structured interview, which asked for their opinions on the assistant, their thoughts on customisation, their shopping experience, and their preference for the main VR sessions. For each full experiment, participants were given a budget of \$40 to buy food and drinks for a movie night consisting of three people.

### 3.5 Measures

- Stress and Anxiety. We used the 2-item 5-point Likert scale stress and anxiety questions from [1], which were collected before and after the study.
- Tele-presence. We used the 7-point Likert scale Slater-Usuh-Steed presence questionnaire [50] to assess the user's sense of tele-presence.
- Co-presence. We used the questionnaire from [51], a 5-point Likert scale questionnaire, which assessed the user's sense of co-presence towards the assistant.
- Social presence. We used the 7-point Likert scale social presence from [39] to assess the user's sense of social presence towards the assistant. This social presence was chosen over a sense of anthropomorphism [52] or trust [20] because it was more appropriate for the VR setting, and there is existing research that found strong relationships between social presence, anthropomorphism, and trust [46, 47].
- Store metrics. We also collected the total amount spent in the store and the number of times users engaged with the assistant during each VR shopping session. As a measure of shopping motivations.

### 3.6 Participants

A total of 27 participants (20 male, 7 female) with a mean age of 23.13 (SD = 6.89, range = 20 to 56) were recruited for this study. The sample size was calculated through a power analysis with 0.8 power, 0.05 alpha, and 0.5 effect size, DF = 26, and total sample size = 27.

## 4 Study Results

### 4.1 Quantitative Analysis

Normality of the non-categorical data types was tested through the Shapiro-Wilk and Q-Q plot test. The paired sample t-test was used when normality was determined; otherwise, Wilcoxon signed-rank test was used. Comparisons were made between customised assistants (C) and non-customised assistants (NC) and the participants' first (F) and second (S) experiences for ordering effect. The process was to first attempt to find statistically significant differences in the VR experience through customisation. If there are no changes, we attempt to find changes through other potential factors.

**4.1.1 VR Presence.** We found a significant difference between C and NC ( $T(26) = 2.27, p = 0.016$ ) for Social Presence. Regarding the overall SUS score, the results yielded no significant difference between C and NC ( $T(26) = -0.29, p = 0.61$ ) nor F and S ( $T(26) = 1.15, p = 0.26$ ). We also could not find any significant difference between C and NC ( $T(26) = 0.68, p = 0.25$ ), F and S ( $T(26) = 0.27, p = 0.79$ ) regarding Co-Presence.

**4.1.2 Stress and Anxiety.** The changes in stress and anxiety were calculated by the difference between the pre- and post-measurements. We could not find any significant results: C and NC stress ( $z = 1.03, p = 0.86$ ), F and S stress ( $z = 1.03, p = 0.33$ ), C and NC anxiety ( $z = 1.51, p = 0.94$ ), F and S anxiety ( $z = 0.905, p = 0.41$ ).

**4.1.3 Amount Spent in Store.** There was a statistically significant increase in the amount spent seen in Table 1. C and NC ( $z = 1.95, p = 0.026$ ), F and S ( $z = 0.336, p = 0.75$ ). A post hoc analysis was taken to see if there were any effects to budgeting from C and NC or F and S. Of the 10 instances of overspending, 6 (60%) were committed with a customised assistant which suggests no visible effects through customisation, while only 3 (30%) instances were committed on the second VR experience. An additional note is that one participant went over budget for both VR experiences; excluding this outlier leads to 5 (56%) instances of overspending with a customised assistant and only 2 (22%) instances of overspending on the participant's second VR experience. These points suggest that either users were not interested in following their budget or that customisation did not help with the function of the assistant. This is supported by the number of interactions.

**4.1.4 Assistant Interactions.** Interactions were measured as the number of questions asked of the assistant (i.e., success prompt with answers from ChatGPT). We could not observe statistically significant differences between C and NC ( $z = 0.35, p = 0.12$ ). There were also no order effects: F and S ( $z = 0.32, p = 0.15$ ).

**4.1.5 Customisation Choice.** First, a chi-squared goodness of fit test was done to identify potential preferences for customisation. There were no statistically significant differences found in the choices of customisation (Age;  $X^2 = 0.862, p = 0.353$ ), (Gender;  $X^2 = 0.862, p = 0.353$ ), (Clothing;  $X^2 = 1.689, p = 0.194$ ), (Skin Tone;  $X^2 = 4.414, p = 0.11$ ). However, the two most chosen assistants were A3 and A13. A3 was chosen 7 times and was classified as Young, Female, Casual, and Light Brown, while A13 was chosen 4 times and was classified as Young, Male, Casual, and Light Brown. Although there were no statistically significant differences, the top two choices

were both Young, Casual, and Light Brown, which may point to a trend. Other choices of assistant were not chosen more than 2 times, except for the default assistant, which was chosen 3 times; in such cases, it was because participants expressed a lack of desire to customise their assistant.

**4.1.6 Demographic Influence.** 2 (7%) participants expressed a preference for the non-customised assistant, 12 (44%) expressed no preference and 13 (48%) expressed a preference for the customised assistant. 2 (7%) of the participants "have bought VR products before / played it frequently," 14 (52%) of the participants "tried it a few times," while 11 (40%) had no prior VR experience. The data on VR experience and preference was interesting, only 2 of the 11 participants which had no prior VR experience preferred the customised assistant while only 1 of the 16 participant with some or more VR experience did not prefer the customised assistant, this suggests a strong correlation between VR experience and preference for customisation which corroborates with the findings in [29] between power and non-power users. However, participants with a preference for customised assistants or otherwise still exhibited the same behaviours in terms of perceived social presence and amount spent. Further analysis of the results with demographic data was not done due to the weak distribution of demographic data. 23 of the 27 participants were aged 18-24 (86%) with 2 participants aged 25-34, 1 aged 35-44, and 1 aged 44+. The gender distribution is also similar, with 20 of 27 participants identifying as male and 7 identifying as female.

### 4.2 Qualitative Analysis

We conducted the Thematic analysis through the 6-phase procedure outlined in [53] for the semi-structured interview data. The themes were identified with a focus on relation to the shopping assistant, how participants felt towards them, and interacting with them. The thematic analysis resulted in the following themes.

1) Users felt the shopping assistant was useful during the shopping experience

The majority of participants mentioned the usefulness of the shopping assistant. Participants made a key distinction between real-life shopping assistants and the VR shopping assistant, in that real-life assistants cannot easily provide detailed information such as the total cost of goods in the shopper's basket or the overall calories.

"I think it was very good that they were able to give me an exact price for my budget. So, I could constantly adjust if I was going over or under budget. Because, especially like in real life shopping, I feel like you can never... You don't quantify until you get to the very end when you're paying for it. So, I thought that was important. And I felt it was really cool that they talked about calories. They mentioned the overall calories of what you're about to purchase" [P13]

"It was good cause I lost track of how much my items were costing every time I had to go back and ask him" [P9]

It is through this distinction that the participant felt the VR shopping assistant was useful. Overall, participants suggest that there is an advantage to VR shopping assistants in their ability to provide information that is user-specific, which real-life assistants cannot. This theme links to the information availability and convenience of

**Table 1: Table of quantitative results, mean (standard deviation) of measures across Customised, Non-Customised, First and Second VR experience. For non-normally distributed data sets, marked with \*, the median was used instead of the mean. Statistically significant results are bold**

| Measures                     |                  | Customised                     | Non-customised | First                   | Second       |
|------------------------------|------------------|--------------------------------|----------------|-------------------------|--------------|
| SUS (Telepresence)           | Mean (SD)        | 2.79 (1.18)                    | 2.83 (0.88)    | 2.88 (0.99)             | 2.74 (1.08)  |
|                              | Paired T-test    | T(26) = -0.29, p = 0.61        |                | T(26) = 1.15, p = 0.26  |              |
| Co-Presence                  | Mean (SD)        | 3.02 (0.52)                    | 2.96 (0.44)    | 3.00 (0.49)             | 2.98 (0.48)  |
|                              | Paired T-test    | T(26) = 0.68, p = 0.25         |                | T(26) = 0.27, p = 0.79  |              |
| Social Presence              | Mean (SD)        | -1.70 (4.58)                   | -3.00 (4.63)   | -1.96 (4.42)            | -2.74 (4.85) |
|                              | Paired T-test    | T(26) = 2.27, <b>p = 0.016</b> |                | T(26) = 1.28, p = 0.105 |              |
| Stress Difference*           | Median (SD)      | 0.00 (0.66)                    | 0.00 (0.65)    | 0.00 (0.76)             | 0.00 (0.52)  |
|                              | Signed Rank Test | z = 0.31, p = 0.17             |                | z = 0.22, p = 0.33      |              |
| Anxiety Difference*          | Median (SD)      | 0.00 (0.66)                    | 0.00 (0.55)    | 0.00 (0.75)             | 0.00 (0.44)  |
|                              | Signed Rank Test | z = 0.40, p = 0.08             |                | z = 0.19, p = 0.41      |              |
| Amount Spent in Store (\$) * | Median (SD)      | 37.67 (9.96)                   | 34.55 (13.17)  | 36.94 (13.51)           | 35.83 (9.34) |
|                              | Signed Rank Test | z = 0.50, <b>p = 0.026</b>     |                | z = 0.07, p = 0.75      |              |
| # of Interactions*           | Median (SD)      | 3 (3.97)                       | 2 (2.39)       | 3 (3.62)                | 2 (2.91)     |
|                              | Signed Rank Test | z = 0.35, p = 0.12             |                | z = 0.32, p = 0.15      |              |

utilitarian shopping motivations mentioned by P-L To et al. [34]; however, customisation was not mentioned when talking about the functionality, as such, links between the cosmetic customisation and utilitarian motivations cannot be drawn.

## 2) Anthropomorphism of the Assistant

Another theme was the perceived anthropomorphism of the assistant. Participants describe the assistant as lacking in emotion, which is negative for anthropomorphism. Many participants expressed that the assistant felt more like a machine or a kiosk due to the lack of locomotion and robotic voice. There is also the point of less stress and anxiety felt due to a lack of emotions, which relates to gratification shopping mentioned in [28], though it is only mentioned by one participant.

“Okay, when I’m talking to the virtual shopping assistant. I think they don’t have any emotions, so I feel less pressured and less stressful when I’m talking with someone without any emotions. I don’t have to think about those factors” [P16]

Overall, the assistant feeling robotic would negatively impact anthropomorphism and social presence, possibly lowering the VR Retail experience. Such comments were directed towards the shopping assistant in general and were not related to the customisation process.

With respect to the customisation process, many participants felt familiarity or friendliness with the assistant they chose, which emphasizes the effects of customisation. Regarding this, participants expressed an increase in friendliness with the assistant they customised, which is positive for anthropomorphism and likely relates to the increase in social presence as well.

“I liked the customised assistant more. I could choose someone more friendly looking/approachable - made me want to interact with them more,” [P21]

“Friendly and familiar face. I don’t know, not familiar but just yeah, if anything it was like, it wasn’t overly familiar, I guess. He’s more friendly I guess” [P10]

The frequency with which human qualities are mentioned in describing the shopping assistant alludes to the importance of anthropomorphism and, by extension, social presence regarding the shopping assistant’s role in the VR Retail experience.

## 5 DISCUSSION

Collating the results reveals notable findings. Regarding (RQ1), there was an increase in social presence through customisation found in the quantitative analysis, this increase happened to both groups with VR experience and groups without which suggests that the customisation method was simple such that technology acceptance had little to no impact yet meaningful enough to have visible effects on the VR experience. This increase in social presence likely signifies an increase in trust and anthropomorphism, as mentioned by [19, 20, 45]. The increase in social presence is also corroborated by theme 2 from qualitative analysis. Participants described the customised assistant with phrases that suggest a higher sense of perceived anthropomorphism. Anthropomorphism has been linked to social presence as both refer to the humanity of the shopping assistant [38, 46, 47]. Additionally, Bailenson et al. [39] clarify that a negative social presence score indicates that users felt the shopping assistant was unconscious and unaware, while a positive score indicates consciousness and awareness, which is a concept closely related to anthropomorphism. While the overall measure of social presence for the assistant was negative, indicating the assistant as unconscious and unaware, sentiments expressed by participants through theme 2. The increase in social presence through customisation supports previous literature on customisation [14, 23, 29]. The negative score can be explained by the sense of incompleteness the shopping assistant gave towards the participants, as previously mentioned, users will tend to believe rather than disbelieve. This means by presenting a humanlike avatar for the users to engage with the expectation that the avatar will move like a human was set and by failing to uphold said expectations the users felt a lower sense of social presence with the assistants, this is also mentioned



by Nowak and Biocca [38] “more anthropomorphic images set up higher expectations that lead to reduced presence when these expectations were not met”. However, despite the increased social presence, there was no increased frequency of interaction with the assistant. This could be due to co-presence. Goffman [54], who coined the term co-presence, described it as individuals becoming uniquely accessible; the lack of change in co-presence means that participants did not feel the assistant was more available to them, regardless of how human the shopping assistant felt. Another possible reason for the frequency of interaction could be the combination of the size of the store and the budget of the shopping task. Because there was a soft budget given to participants, in that it was expressed but not enforced, participants did not have much leeway in terms of the number of products purchased. This, in conjunction with the small store, meant that there wasn't much need to communicate with the shopping assistant. As such, potential differences between the frequency of interactions with customised and non-customised assistants could not be properly expressed within the experiment. The lack of effect on the frequency of interactions suggests a separation between the social and functional aspects of the assistant, relating to the hedonic and utilitarian aspects of shopping motivations.

This leads to RQ2, where there was a statistically significant increase in expenditure when users were allowed to customize their virtual assistants. The increased amount spent can be attributed to increased social presence, as it ultimately relates to trust in the shopping assistant and purchasing intention [17, 21]. As mentioned by Komiak et al. [17], customisation increases trust in a recommendation agent. The concepts of social presence, anthropomorphism, and trust relate strongly to each other, given their known relationships [20, 45]. The increase in purchase amount also corroborates this, as Gefen [21] identified the relationship between trust and purchase. Overall, there is a well-supported relationship between customisation and social presence and, by extension, the purchasing amount found in this study.

As for RQ3, there was no conclusive data that supported the idea that a customised assistant had an effect on the VR experience for users' utilitarian or hedonic shopping motivations. The lack of relationships towards utilitarian shopping motivations could be explained as the customisation being purely cosmetic and, therefore, not impacting the ability of the assistant to provide shopping assistance. Though some participants also mentioned that the lack of humanity in a shopping assistant benefited their interactions, alluding to the lack of sociality and gratification in shopping, it was not a commonly expressed which leaves it inconclusive. As Lucia-Palacios et al. [55] mention the existence of stress due to crowded retail environments, the exclusion of other humans or human-like entities could help mitigate this form of stress.

Although a shopping task was introduced with a budget to help induce stress, similar to the method from [1], where stress was induced during the task. It is clear from the results of this study that the shopping task did not induce enough stress and that, instead, participants found the VR experience relaxing. This meant that instead, data would be found through a decrease in the starting stress and anxiety. However, the starting point of participants' stress and anxiety was generally the lowest possible before any VR experiences, which meant that decreases in stress and anxiety could not be

measured. The only point where participants expressed fulfillment of their shopping motivations was in theme 1, where participants mentioned that the utility of the shopping assistant benefited their shopping experience, potentially relating to information availability and cost saving of utilitarian shopping motivations or the value shopping in hedonic shopping motivations. However, this utility was independent of customisation and therefore irrelevant to the research question. The study found that customisation can be a useful tool to enhance the VR Retail experience when applied to virtual avatars. However, there is also a possibility that these findings can be generalised towards the larger VR space, as much of the related work and subsequent research questions are predicated on non-retail VR literature or non-VR literature.

## 6 LIMITATIONS AND FUTURE WORK

One limitation of this study that must be considered is the homogeneity of the participants as there are known gender-based differences relating to shopping motivations [56], large majority of the participants were male. This impacts the generalisability of the findings and should be considered as a point for potential improvement. Future work should aim to involve balanced genders across different age groups for in-depth analysis or even include personality types, which are also known to affect shopping behaviour and shopping motivations [57, 58]. Another limitation is the limited movement of the shopping assistant, as our shopping assistant is fixed in one location. Future work should explore adding more anthropomorphic behaviours to the assistant, as the addition of locomotion would significantly increase the intelligent virtual assistant's perceived reliability and spatial presence, as well as the realism of the avatar [44].

During the experimental procedure, participants' opinions were mixed regarding how they would like the assistant to be presented. The opinions can be categorised into omnipresent/physically bound, which refers to the activation range of the assistant, and life-sized/fairy-sized/digital, which refers to the appearance of the assistant. Research could be done to understand the benefits each option can bring to the VR Retail experience. Finally, we had limited customisation choices, which only encompassed cosmetic changes; further research should be done into the customisation range and its subsequent effects on VR retail. Different customisation options could be explored, such as functional changes regarding the assistant's recommendations, their speech modality, range of locomotion, or the form in which the assistant is presented. Which could target the utilitarian aspect of shopping motivations.

## 7 CONCLUSION

This paper explained the potential benefits that customisation of shopping assistants could provide VR Retail experiences and then conducted a study that investigated these claims. The results of this study strongly suggest that customisation of a VR shopping assistant can enhance the VR Retail experience. This is shown through the increase in the perceived social presence towards the shopping assistant as well as the increase in the amount spent within the VR store. Overall, the results of this study provide insights into the role of VR shopping assistants and the benefits of customisation in VR Retail. Such findings could lead to potentially

interesting results when investigated in other VR fields, such as mental health, education, or entertainment.

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